

Bit Manipulation

Bitwise operation

① Binary AND or Bitwise AND

② Binary OR

③ Binary XOR

④ One's complement $\sim$ (not operator)

⑤ Binary left shift $\ll$

⑥ Binary Right shift $\gg$

$$\begin{array}{r} (11)_{10} \rightarrow 1011 \\ (9)_{10} \leftarrow + 1001 \\ \hline 2010 \\ \underline{20} \end{array}$$

Bit $\rightarrow$ Binary Digit

X	Y	$x \& y$	$x y$	$x \wedge y$
0	0	0	0	1
0	1	0	1	0
1	0	0	1	0
1	1	1	1	1

XOR

$$x \oplus y = x\bar{y} + \bar{x}y$$

x	$\sim x$
0	1
1	0

$$\begin{array}{r} 10 \& 12 = 1010 \\ \& 1100 \\ \hline (1000)_2 = 8 \end{array}$$

$$\begin{array}{r} 10 | 12 = 14 \\ \vee 1010 \\ \vee 1100 \\ \hline (1110)_2 = 14 \end{array}$$

$$\sim (10110110)_2 = (01001001)_2 \xrightarrow[+1]{2^{nd} \text{ comp}} (01001010)_2$$

$\xrightarrow{\text{1's comp}}$

$$(10111000)_2 \xrightarrow{2^{nd} \text{ comp}} (01001000)_2$$

n bit

the range of +ve integers = $[0, 2^{n-1}-1]$

↓ negative numbers

n=3 = $[-3, 3]$

0	0	0	→ +0
0	0	1	→ +1
0	1	0	→ +2
0	1	1	→ +3
1	0	0	→ -3
1	0	1	→ -2
1	1	0	→ -1
1	1	1	→ -0

$$\begin{aligned} +0 &= 000 \\ -0 &= 111 \end{aligned} \quad \left. \vphantom{\begin{aligned} +0 &= 000 \\ -0 &= 111 \end{aligned}} \right\} 0$$

$$\text{Range [negative + positive]} = [-2^{n-1}, 2^{n-1}-1]$$

↓
[1's complement]

Positive negative
2nd complement

3 bit integer

0	0	0	→ 0
0	0	1	→ 1
0	1	0	→ 2
0	1	1	→ 3
1	0	0	→ -4
1	0	1	→ -3
1	1	0	→ -2
1	1	1	→ -1

3 bit

$$(-4)_{10}$$

$$(4)_{10} = (100)_2$$

↓
2nd comp X

MSB

MSB = 0 → +ve
1 → -ve.

bit size = 3

$$(-3)_{10} = (?)_2$$

$$(3)_{10} = (011)_2 \xrightarrow{2^{nd}} (101)_2$$

↓ 1's comp

$$\sim(3)_{10} = 100$$

$(101)_2$ its negative representation of 3 i.e -3

$$\text{Range} = [-2^{n-1}, 2^{n-1}-1]$$

for 4 bit
Range → $[-8, 7]$

Right shift

$$(25)_{10} \gg 2$$

$$16 + 8 + 1$$

$$(25)_{10} = (0011001)_2 \gg 2$$

$$(25)_{10} \gg 2 = 6$$

$$25 / 4 = \lfloor 6.25 \rfloor$$



$$= 6$$

↓ ↓
4 2

$$\text{num} \gg \text{shift} = \lfloor \text{num} / 2^{\text{shift}} \rfloor$$

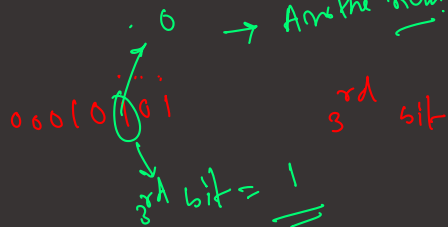
→ Answer num. (new num)

Q any Number will be given, finds

(i) Get i^{th} bit

(ii) Set i^{th} bit → new number [i^{th} bit = 1]

(iii) clear i^{th} bit. [i^{th} bit = 0]
→ new number



3rd bit

00010101

00000100

00000100 $\gg 2$

4th bit

00010101

00001000

(00000000)₂ = (0)₁₀

bit = 0

bit mask

i^{th} bit = ?

Magical Num = $1 \ll (i-1)$

Given Number & Magical Number = Result $\gg (i-1)$

0
bit 0
1
bit 1

day-8 integer
 num = 10100100
 2
 bin 10
 (or) 10101100
 set(3)
 10100100
 00001000 ← bit mask (magical num).
 1 << i

$$(00000001)_2 \ll 3$$

$$= (00001000)_2$$

Set bit = num | (1 << i)

Clean ith bit
 10011100
 0 2nd bit ← 3rd position
 and (2) 11111011 ← bit mask.
 bit mask ~ (1 << i)

$$\sim [(00000001)_2 \ll 2]$$

$$\sim (00000100) = 11111011$$

$$\begin{array}{r} 10010010 \\ 11101111 \\ \hline 10000010 \end{array}$$

Clean last i bit

num = 10011011
 ans → 10010000

i = 4

ans = num & [~0 << i]

num = 10011011
 and (2) 11110000 ← bit mask.

$$(\sim 0) = -1$$

$$= 11111111$$

bit mask = (~0) << i

$$-1 \ll 4 = 11110000$$

